Math 8 Accelerated
Understanding and Solving Standards



— SINE TIMORE AUT FAVORE —

ST. GEORGE'S SCHOOL

Name : _____

Teacher's name: _____

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1 Rational Numbers

No calculators

Emerging Level

- Convert mixed fractions to improper fractions
- Convert improper fractions to mixed fractions
- Convert decimals to fractions
- Evaluate simple arithmetic operations with rational numbers

Developing Level

- Identify rational numbers
- Classify rational numbers
- Compare rational numbers

Proficient Level

- Evaluate order of operations with rational numbers

Extending Level

- Model and solve application problems, including multi-step
- Simplify complex fractions

- Solve unfamiliar problems
- Answer open-ended problems with explanation

2 Linear Equations and Inequalities

No calculators

Emerging Level

- Write simple algebraic expressions or equations from words
  - The sum of twice x and y
  - H is the difference of x and half y
- Solve one-step linear equations
- Solve two-step linear equations
- Evaluate expressions with simple substitutions
- Validate answers by left and right side check

Developing Level

- Solve linear equations with multiple instances of x
- Solve linear equations with distributive law - keeping to integer coefficients

Proficient Level

- Solve linear equations with fractions and decimals in different contexts
  - with common denominators
  - without common denominators
  - with one to two decimal places
  - etc...

Extending Level

- Model and solve application problems with linear equations
- Solve linear inequalities and write answers as an inequality and model on a number line

- Solve unfamiliar problems
- Answer open-ended problems with explanation

3 Roots, Powers, and Exponent Laws

No calculators

Emerging Level

- Represent numbers using powers and exponents
- Evaluate simple expressions with exponents
- Evaluating roots of numbers with rational answers
- Represent a number as a product of primes
- Simplify expressions with exponent laws (simple)

Developing Level

- Evaluate simple order of operations with exponents
- Compare numbers with exponents
- Estimate roots of numbers with irrational numbers
- Understand radicals and roots as inverse operations of exponents

Proficient Level

- Apply multiple exponent laws to simplify an expression
- Evaluate more complicated order of operations with exponents
- Evaluate roots of larger numbers by prime factorization

Extending Level

- Apply multiple exponent laws to simplify a complex expression
- Simplify an expression using a change of base and then exponent laws
- Simplify more complicated expressions with negative exponents
- Solve for the unknown using inverse operations on exponents
  - ie. if $x^2 = 4$, then $x = \pm\sqrt{4} = \pm 2$

- Solve unfamiliar problems
- Answer open-ended problems with explanation

4 Pythagorean Theorem, Surface Area and Volume

Calculators Required

Emerging Level

- Identify solids by name given sketch or net diagram
- Sketch a net diagram of a given solid
- Sketch and label the sides of a given solid with measurements
- Find the surface area given a net diagram with measurements

Developing Level

- Find surface area and volume of rectangular prisms
- Find surface area and volume of cylinders
- Solve right-triangles with the Pythagorean Theorem
- Verifying if a triangle is a right triangle

Proficient Level

- Find surface area and volume of triangular prisms
- Find surface area and volume of cones
- Find surface area and volume of spheres
- Find volume of composite solids

Extending Level

- Find surface area of composite solids
- Find a missing measurement given the volume
- Find a missing measurement given the surface area

- Solve unfamiliar problems
- Answer open-ended problems with explanation

5 Linear Relations

Emerging Level

- Plot points on a grid (integral coordinates)
- Understand linear patterns
  - State next terms
  - State common difference
- Use provided linear equation to find information
  - Find t given n
  - Find n given t
- Write equations for linear patterns

Developing Level

- Complete a table of values for a given linear relation and then graph ($y = mx+b$)
- Sketch linear equations
  - Use slope-intercept form: $y = mx + b$
- Sketch horizontal and vertical lines

Proficient Level

- Sketch linear equations
  - Use general form: $Ax + By = C$
  - Apply integer coefficients
- Write linear equations of linear graphs (integer y-intercept)
- Solve for coordinates given an equation
  - Find missing x in $(___, y)$
  - Find missing y in $(x, ___)$

Extending Level

- Write linear equations with rational y-intercept
- Sketch linear equations with rational coefficients
- Use linear equations to solve linear application problems
- Model a linear relation problem through a table of values and graph
- Use interpolation and extrapolation to find/estimate values

- Solve unfamiliar problems
- Answer open-ended problems with explanation

6 Polynomials

Emerging Level

- Classify polynomials
  - Identify monomials, binomials, and trinomials
  - Identify leading coefficients and terms
  - Determine the degree
- Write expressions modelled by algebra tiles
- Simplify expressions by collecting like terms

Developing Level

- Add and subtract polynomials
  - Solve horizontally and vertically
- Multiply simple polynomials
  - Multiply a monomial by a monomial

Proficient Level

- Multiply complex polynomials
  - Multiply a monomial by a binomial
  - Multiply a monomial by a trinomial
- Divide polynomials by a monomial

Extending Level

- Apply multiple operations using BEDMAS
- Multiply polynomials beyond just a monomial
  - Evaluate expressions like $(2x + 3)(4x - 5)$
- Solve area problems involving polynomials

- Solve unfamiliar problems
- Answer open-ended problems with explanation

7 Rates, Proportions, and Similarity

Calculators Required

Emerging Level

- Simplify ratios, rates, and fractions
- Find unit rates
- Convert between fractions, decimals, and percentages
- Identify enlargements and reductions
- Work with scale factors, objects, and images

Developing Level

- Simplify ratios with simple unit conversions
- Solve for an unknown in a proportion equation
- Solve for missing side lengths in simple similar polygons and triangles

Proficient Level

- Solve percent problems
- Solve simple word problems involving proportions or percentages
- Justify similarity in polygons or explain why they are not similar
- Solve for missing side lengths in similar polygons and triangles using algebra

Extending Level

- Solve complex word problems involving proportions
- Solve compound percent problems and their applications
- Solve complicated cases involving similar polygons
- Solve word problems requiring similar polygons or proportions

- Solve unfamiliar problems
- Answer open-ended problems with explanation

8 Statistics and Probability

Calculators Required

Emerging Level

- Define different measures of central tendency
- Identify measures of the center given a data set
  - Determine the mean, median, and mode
- Interpret data from simple graphs

Developing Level

- Justify properties of graphs
- Identify misrepresentations in graphs with explanation

Proficient Level

- Calculate simple probabilities
  - Analyze scenarios involving dice, spinners, cards, and marbles
- Identify independent and dependent events
- Find probabilities of multiple events

Extending Level

- Solve for missing data values given the mean, median, mode, etc.
- Solve complicated probability scenarios

- Solve unfamiliar problems
- Answer open-ended problems with explanation